

Original papers

Cattle verification with YOLO and cross-attention encoder-based pairwise triplet loss

Niraj Kumar^a, Anshul Sharma^b, Abhinav Kumar^d, Rishav Singh^c, Sanjay Kumar Singh^{a,*}^a Indian Institute of Technology (BHU), Varanasi, Department of Computer Science and Engineering, Varanasi, 221005, Uttar Pradesh, India^b National Institute of Technology, Patna, Department of Computer Science and Engineering, Patna, 800001, Bihar, India^c Indian Institute of Technology, Patna, Department of Computer Science and Engineering, Varanasi, 801106, Bihar, India^d Indian Institute of Technology, Ropar, School of Artificial Intelligence and Data Engineering, Ropar, 141001, Punjab, India

ARTICLE INFO

Keywords:

Cattle verification
Biometric identification
Deep learning
Computational efficiency
YOLO v10

ABSTRACT

Cattle identification through biometrics has increasingly relied on non-invasive methods, where facial and muzzle images are commonly used. While facial images alone offer valuable features, they can be sensitive to variations in lighting, pose, and occlusion, leading to inconsistent results. Similarly, relying solely on muzzle images, though distinct, may not provide sufficient discriminatory power for accurate identification in all cases. Combining both facial and muzzle features offers a more comprehensive solution, overcoming the limitations of using either feature independently. Recent advancements in deep neural networks have significantly influenced this field, but achieving an optimal balance between model accuracy and computational efficiency remains a challenge. In this paper, we present a novel approach by introducing a **Cross-Attention Encoder in combination with a Pairwise Triplet Loss function for cattle verification**. This method processes face and muzzle images in parallel, enhancing the integration of features from both inputs. The cross-attention mechanism enables the model to focus on the most relevant regions, improving feature alignment and discrimination. With a compact architecture of only 0.6 million parameters, the encoder effectively captures essential features from both the face and muzzle, ensuring precise cattle verification without excessive computational demands. Our approach achieved a testing accuracy of 93.67%, with an **average inference time of 35ms** per sample, demonstrating the model's efficiency. These findings highlight the strength of our attention-based method in delivering high accuracy and computational performance for cattle verification tasks.

1. Introduction

The emerging field of animal biometric systems aims to develop methodologies for recognizing and detecting various species, their traits, behaviors, and biometric characteristics using pattern recognition (Kuhl and Burghardt, 2013). Traditional cattle recognition methods, such as ear tags and ear tattoos, are invasive and suffer from fraud and duplication, leading to potential threats to animal welfare and health problems for the cattle (Noviyanto and Arymurthy, 2013).

Existing cattle identification technologies face significant challenges in accuracy, efficiency, and scalability. Traditional methods are prone to wear and tear, loss, and fraudulent manipulation. Additionally, these methods often involve invasive procedures that can compromise

the well-being of the animals. Manual inspection processes are labor-intensive, time-consuming, and susceptible to human error, further exacerbating inefficiencies. In contrast, modern deep learning-based technologies offer transformative potential by enabling automated, non-invasive, and highly accurate cattle identification systems. These systems can significantly reduce fraud, improve traceability, and streamline livestock management. However, the transition from traditional methods to advanced technologies necessitates addressing the gaps in robustness, scalability, and adaptability to diverse environmental conditions, which remain persistent in existing systems.

Cattle recognition is pivotal in deep learning, offering valuable data for improving cattle management and care (Kumar et al., 2016). As cattle recognition systems focus more on recognizing and differentiating

* Corresponding author.

E-mail addresses: nirajkumar.rs.cse22@itbhu.ac.in (N. Kumar), anshul.cs@nitp.ac.in (A. Sharma), abhinavk@iitrpr.ac.in (A. Kumar), rishav.singh@iitp.ac.in (R. Singh), sks.cse@iitbhu.ac.in (S.K. Singh).<https://doi.org/10.1016/j.compag.2025.110223>

Received 7 September 2024; Received in revised form 18 February 2025; Accepted 1 March 2025

Available online 14 March 2025

0168-1699/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

species based on unique features, some traditional cattle recognition methods are prone to issues, making them insufficient for livestock protection. The adoption of modern techniques has brought about revolutionary solutions to a range of challenges within the livestock industry, particularly in cattle management. These techniques have successfully addressed longstanding problems, including the potential for cattle swapping, fraudulent insurance claims, and disputes over ownership. One of the key contributions of these advancements lies in their ability to provide a robust and transparent system for cattle identification and verification. Moreover, they play a pivotal role in ensuring the safety of animal-derived food products and offer a means of validating the various stages of cattle production, processing, and exportation.

The need for a new cattle identification system stems from the limitations of current approaches and the growing demand for precise and efficient livestock management solutions. Traditional methods lack the ability to uniquely and reliably identify individual animals under varying conditions, such as poor lighting, occlusions, or environmental noise. This creates an urgent requirement for a system that can operate effectively in real-world scenarios while addressing issues like motion blur, pose variation, and incomplete data. Our study fills these gaps by introducing a novel approach by utilizing deep learning based feature extraction method. This framework capitalizes on the complementary features of facial and muzzle biometrics to enhance verification accuracy. By doing so, it not only addresses the limitations of traditional systems but also sets the stage for advancements in smart livestock management, ensuring seamless integration into existing agricultural workflows.

At the heart of these technological innovations is the application of deep learning in biometrics, a cutting-edge approach that leverages the unique and distinctive features of individual cattle through the extraction and analysis of both facial and muzzle characteristics. The face, including the eyes, forehead, and other structural features, combined with the distinct texture and shape of the muzzle, provides a comprehensive biometric signature for each animal. The muzzle, which includes the nostrils, upper lip, and surrounding skin texture, is particularly unique, much like a human fingerprint, while the facial features add an additional layer of distinctiveness (Noviyanto and Arymurthy, 2013; Kumar et al., 2016). Together, these biometrics offer a robust and reliable method for cattle verification, as illustrated in Fig. 1.

This approach offers several advantages, foremost among them being heightened security and accuracy in cattle verification. By utilizing advanced image processing techniques, it becomes possible to establish an unambiguous link between a cattle's identity and its corresponding biometric traits, effectively eliminating the possibility of mistaken identity or cattle swapping. One of the immediate benefits of implementing deep learning in cattle management is the prevention of false insurance claims. With precise and reliable recognition mechanisms in place, insurers can confidently verify the authenticity of claims related to cattle incidents, mitigating the risk of fraudulent activities. Additionally, the use of deep learning contributes to the resolution of ownership disputes, a common issue in the cattle industry. By relying on irrefutable biometric data, disputes over cattle ownership can be efficiently resolved, reducing legal complexities and potential financial losses. Furthermore, it plays a pivotal role in ensuring the safety of animal-derived food products, addressing a critical concern in the livestock and food industries. Deep learning serves as a valuable tool for validating various aspects of cattle production, processing, and exports. It offers a means of monitoring and verifying the conditions in which cattle are raised, ensuring compliance with ethical and environmental standards.

To address the challenges of accurate cattle verification under diverse conditions, the Siamese triplet model (Schroff et al., 2015) has proven to be particularly well-suited, excelling in learning distinctive features from images, even when the differences between individual

animals are subtle. This model comprises three identical neural networks, with the primary objective of minimizing the distance between the embeddings of the anchor image and the two cattle images when they pertain to the same animal and maximizing this distance otherwise. This capability is critical for reliable cattle verification, a key aspect in monitoring health, productivity, and breeding in agriculture. Furthermore, Siamese triplet models offer several advantages, including reduced data labeling requirements, adaptability to varying image dimensions and environmental conditions, and enhanced resilience to noise, distinguishing them from other deep learning models. To solve the issue of further improving feature integration and verification accuracy, we propose a novel approach that combines a Cross-Attention Encoder with a Pairwise Triplet Loss function, which we outline in detail in Section 3.

Finally, we describe the contributions of this paper as follows:

1. A novel approach to cattle verification is introduced, utilizing face and muzzle images in combination with a pairwise triplet loss architecture. This method addresses the limitations found in other state-of-the-art models, and, to the best of our knowledge, cattle verification in this manner has not been previously explored.
2. A Cross-Attention Encoder model is developed with minimal learning parameters, ensuring computational efficiency while maintaining high performance. The cross-attention mechanism enables the model to dynamically weigh the importance of different parts of the images, resulting in more precise and robust feature fusion. This leads to better discrimination between cattle, even in challenging scenarios such as varying lighting conditions, partial occlusions, or diverse poses.
3. Muzzle images are extracted from frontal cattle images using the state-of-the-art YOLO v10 architecture, alongside various augmentation and transformation techniques to introduce diversity in the dataset.
4. The proposed pairwise triplet loss function effectively leverages both face and muzzle features. By utilizing Manhattan Distance as the metric, this modification brings several advantages, such as reduced sensitivity to irrelevant dimensions, enhanced feature independence, and better accommodation of sparse data.

The paper is organized as follows: In Section 2, we review related works on cattle verification and Triplet loss, while Section 3 elaborates on our proposed Methodology. Section 4 presents our experimental results and discussion, and we conclude in Section 5.

2. Literature review

In recent years, animal biometrics has gained significant traction in research, particularly within livestock management. This growing interest is driven by the need for scalable, cost-effective, and non-invasive solutions for recognizing and monitoring livestock. A notable contribution in this area is from Duong et al. (2018), who introduced an efficient cattle face recognition model, outperforming other convolutional neural networks in terms of accuracy. Their approach extends beyond cattle, with applicability to species like pigs and sheep. Similarly, Zhao and Lian (2022) presented a compact loss function that enables cattle recognition from various angles in natural environments, offering significant potential for practical livestock management. Shen et al. (2020) also explored convolutional neural networks for individual identification of dairy cows, highlighting their utility in precision livestock farming. Xu et al. (2022) further advanced cattle recognition through their CattleFaceNet, which integrates adaptive algorithms to enhance recognition accuracy. Their work resulted in the development of SDBCN, leveraging Dense Block and Capsule Networks for improved performance. Xu et al. (2024) proposed an embedding enhancement and optimization approach to improve cattle

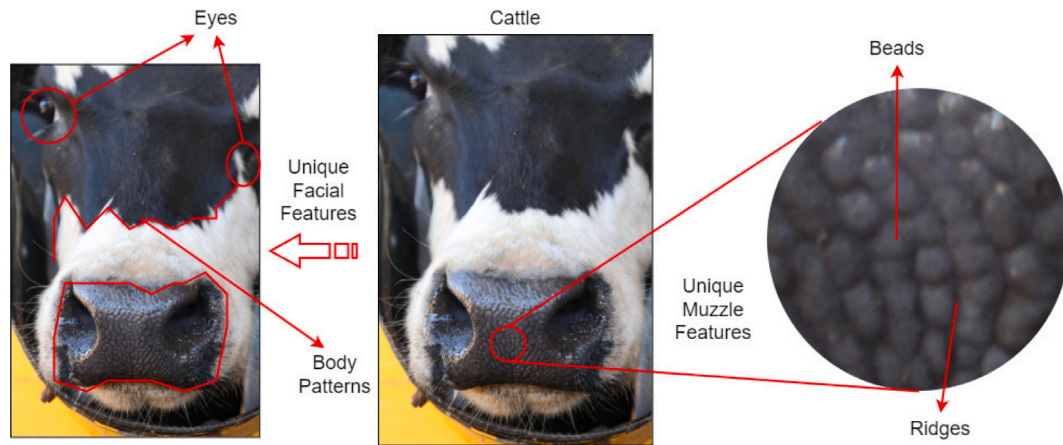


Fig. 1. Illustration of distinctive facial and muzzle features in cattle used for individual recognition. The left section highlights key facial features, including the eyes, body patterns, and overall unique facial contours, and on the right, a magnified view of the muzzle reveals specific beads and ridges, i.e., microscopic details unique to each cattle's muzzle pattern.

face recognition under uncontrolled conditions. Their method integrates deep feature optimization to address variations in lighting, pose, and background, thereby boosting recognition accuracy in real-world scenarios. Li et al. (2024) introduced a multi-feature decision layer fusion technique for cattle identification, improving recognition accuracy by effectively integrating complementary feature representations. Similarly, Dulal et al. (2025) proposed a Multi-Head Attention Feature Fusion (MHAFF) framework that combines CNN and Transformer-based embeddings, enhancing cattle identification by capturing both local and global feature dependencies. Bhole et al. (2022) also made strides by employing computer vision and deep learning techniques, creating an automated system that recognizes individual Holstein cattle from side views. Dubourvieux et al. (2023) contributed by introducing an unsupervised multi-domain adaptation approach for Holstein cattle re-identification, improving robustness across varying environmental conditions. Collectively, these studies underscore the growing role of computer vision and deep learning in livestock management.

Siamese Networks have emerged as pivotal in applications such as face recognition, image classification, and biometric identification, which closely aligns with the dual-input architecture used in cattle verification. Pinto et al. (2021) addressed biometric model security by reformulating the Secure Triplet Loss to ensure template non-linkability, a concept that resonates with the pairwise triplet loss used in this work for cattle verification. Liu et al. (2020) introduced CAL-HFR, leveraging attribute discriminative information to enhance face matching performance, while Ren and Xue (2020) combined YOLOv3 with Double Triplet Networks to improve multi-camera tracking, highlighting the versatility of triplet loss in recognition systems. Similarly, Ma et al. (2021) optimized intra-class and inter-class margins using NMM loss for deep open-set image classification, while Yu et al. (2020) enhanced feature representation with DCNN and novel loss layers. Xie et al. (2022) introduced a triplet loss method for face recognition, improving accuracy by integrating class-pair margins and multistage outlier suppression. Mahato and Neethirajan (2024) demonstrated the potential of facial recognition technology for sustainable dairy practices, integrating artificial intelligence to enhance biometric identification in dairy management. These advancements reflect the evolving role of loss functions in improving recognition accuracy, paralleling the pairwise triplet loss used in this work for cattle verification with face and muzzle images.

To enhance the accuracy of cattle verification, object detection is a crucial step for extracting regions of interest, such as the muzzle. Lin et al. (2018) tackled the challenge of class imbalance in object detection by introducing focal loss, leading to state-of-the-art performance. Ren et al. (2021) proposed Region Proposal Networks, which significantly

enhanced object detection accuracy by efficiently generating region proposals. Building on these advancements, Wang et al. (2022) introduced YOLOv7, a real-time object detection system that achieved impressive results. Lu et al. (2023) proposed a novel algorithm based on locating key areas, contributing to the identification of cattle by refining the region of interest. Girshick et al. (2014) further contributed to the field by enhancing object detection performance with convolutional neural networks and a supervised pre-training/domain-specific fine-tuning approach. These methods provide a foundation for the object detection techniques employed in this research for extracting relevant regions from cattle images.

In this work, we aim to overcome the limitations of previous models by proposing a novel approach to cattle verification that combines both face and muzzle images using a Cross-Attention Encoder and pairwise triplet loss, along with the use of YOLO v10 for accurate muzzle extraction, thereby enhancing recognition accuracy and vitality in real-world scenarios.

3. Proposed methodology

This section provides a concise overview of the methods used, beginning with data acquisition, followed by preprocessing, and covering all the techniques involved in the proposed loss function.

3.1. Data acquisition

The dataset, meticulously prepared and made publicly available by Shojaeipour et al. (2021), includes an extensive collection of 2900 images representing a diverse group of 300 cattle. Facial images of cattle were collected at the University of New England's Tullimba Research Feedlot during an Induction Day event in February 2019. There is a total of 2900 high-resolution images (4000×6000 pixels, RGB mode) captured using a Canon D800 camera, with approximately 9.24% of images excluded due to poor quality. cattle group comprises various breeds, including *Bos taurus* beef breeds such as Simmental, Hereford, Charolais, and Angus. Although we identified two additional publicly available datasets, we chose to use this dataset for our experiments due to its superior image quality.

3.2. Data preprocessing

This subsection provides detailed information about the preprocessing steps undertaken to prepare the data for subsequent processing in the proposed model.

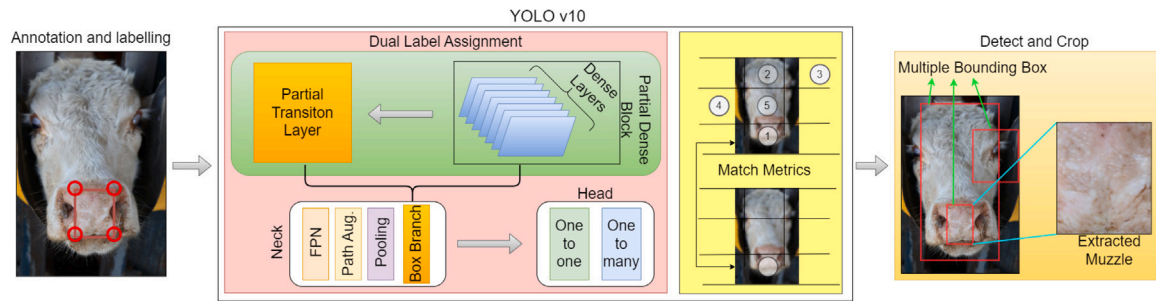


Fig. 2. YOLO v10 muzzle detection and cropping from cattle face images. The central section including the Backbone and layers, which are responsible for feature extraction and detection refinement. The match metrics at the bottom ensure accurate localization.

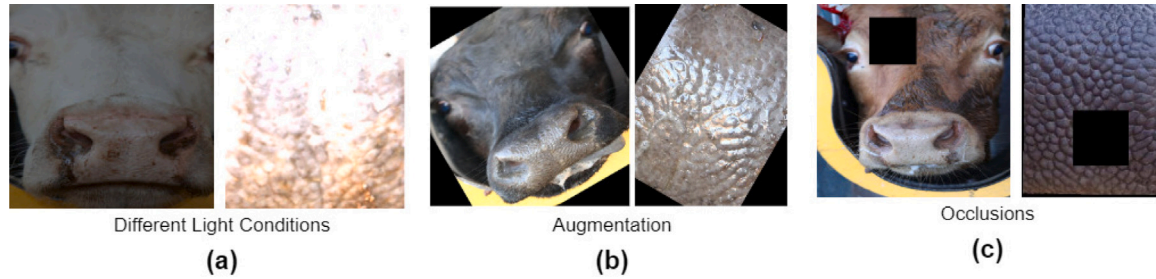


Fig. 3. This figure demonstrates the application of various augmentations and transformations to cattle face and muzzle images to introduce diversity in the dataset. (a) Images are captured under varying lighting scenarios to simulate real-world conditions (b) Different geometric transformations, including rotations and flips, are applied to the muzzle images, enhancing the dataset's variability (c) Intentional obstructions are introduced, covering portions of the face and muzzle, simulating conditions where parts of the animal are obscured.

3.2.1. Muzzle detection and cropping

Accurate detection and extraction of the muzzle region is a crucial step in cattle verification, which can be accomplished by object detection algorithms like YOLO (Redmon et al., 2016; Bochkovskiy et al., 2020). To tackle this challenge, we utilized the YOLO v10 object detector (Reis et al., 2024) and applied transfer learning to fine-tune the network weights for effective muzzle detection. YOLO v10 introduces significant improvements over its predecessors and other conventional object detection methods. Unlike earlier YOLO versions and traditional techniques that require multiple passes through an image to identify objects, YOLO v10 divides the input image into a grid and, in a single pass, predicts bounding boxes and class probabilities for each grid cell (Wang et al., 2024).

The core architecture of YOLO v10, as depicted in the middle section of Fig. 2, includes key components such as the backbone, neck, and head, which work together to optimize object detection performance. The backbone extracts essential features from the input image through convolutional layers. The neck, enhanced by the introduction of a Partial Transition Layer and Path Aggregation Network (PAN) (Liu et al., 2018), consolidates multi-scale feature maps, enabling better detection of objects of varying sizes, such as the muzzle. The dense layers in the Partial Dense Block further refine these features, ensuring precise localization and classification. The head employs a dual label assignment strategy, combining “one-to-one” and “one-to-many” assignments to optimize bounding box predictions, allowing the model to handle multiple overlapping bounding boxes effectively. This ensures that the final bounding boxes, which are non-overlapping, accurately encapsulate the muzzle region.

Compared to other detection methods like Faster R-CNN (Ren et al., 2016), YOLO v10 offers superior speed due to its single-shot nature, which eliminates the need for region proposal networks and reduces inference time. Each grid cell in YOLO v10 generates several bounding box predictions, complete with confidence scores and class probabilities, ensuring thorough coverage. The model further enhances accuracy through non-maximum suppression, which removes overlapping boxes to improve object detection precision. After the model is successfully

trained on muzzle images, the muzzle is cropped to extract specific regions, as shown in the right portion of Fig. 2. Following this, data augmentation and additional image transformations are applied to expand the dataset and enhance the model's robustness. A brief illustration of the YOLO v10 process for muzzle detection and cropping is shown in Fig. 2.

3.2.2. Data augmentation and transformation

In our research, we applied some augmentation and transformation techniques to the face and muzzle images to enhance the robustness of our model. Fig. 3(a), shows brightness adjustment to vary the lighting conditions of the images, creating three versions with decreased, unchanged (original), and increased brightness levels. This helps the model learn to handle different lighting scenarios. Fig. 3(b), illustrates rotation to simulate variations in the orientation of the face and muzzle, generating images rotated by -30 , -15 , and 30 degrees. This allows the model to become invariant to small changes in angle. Fig. 3(c), shows occlusions by adding random black squares to different regions of the images. This technique trains the model to recognize the muzzle or face even when parts of it are obscured. Together, these transformations create a diverse set of training examples that improve the model's ability to generalize and accurately verify the cattle under various conditions.

3.3. Pairwise triplet loss

The pairwise triplet loss, a novel variant of the traditional triplet loss (Hoffer and Ailon, 2018), introduces image similarity learning (Schroff et al., 2015). In the context of cattle verification, this method refines the model's ability to distinguish between individual cattle by leveraging both facial and muzzle features. Specifically, the anchor (A) represents a pair of cattle face and muzzle images from a single cattle, the positive (P) is another pair from the same cattle, and the negative (N) is a pair from a different cattle. The loss function aims to ensure that the distance between the anchor and positive pairs is minimized while the distance between the anchor and negative pairs is maximized.

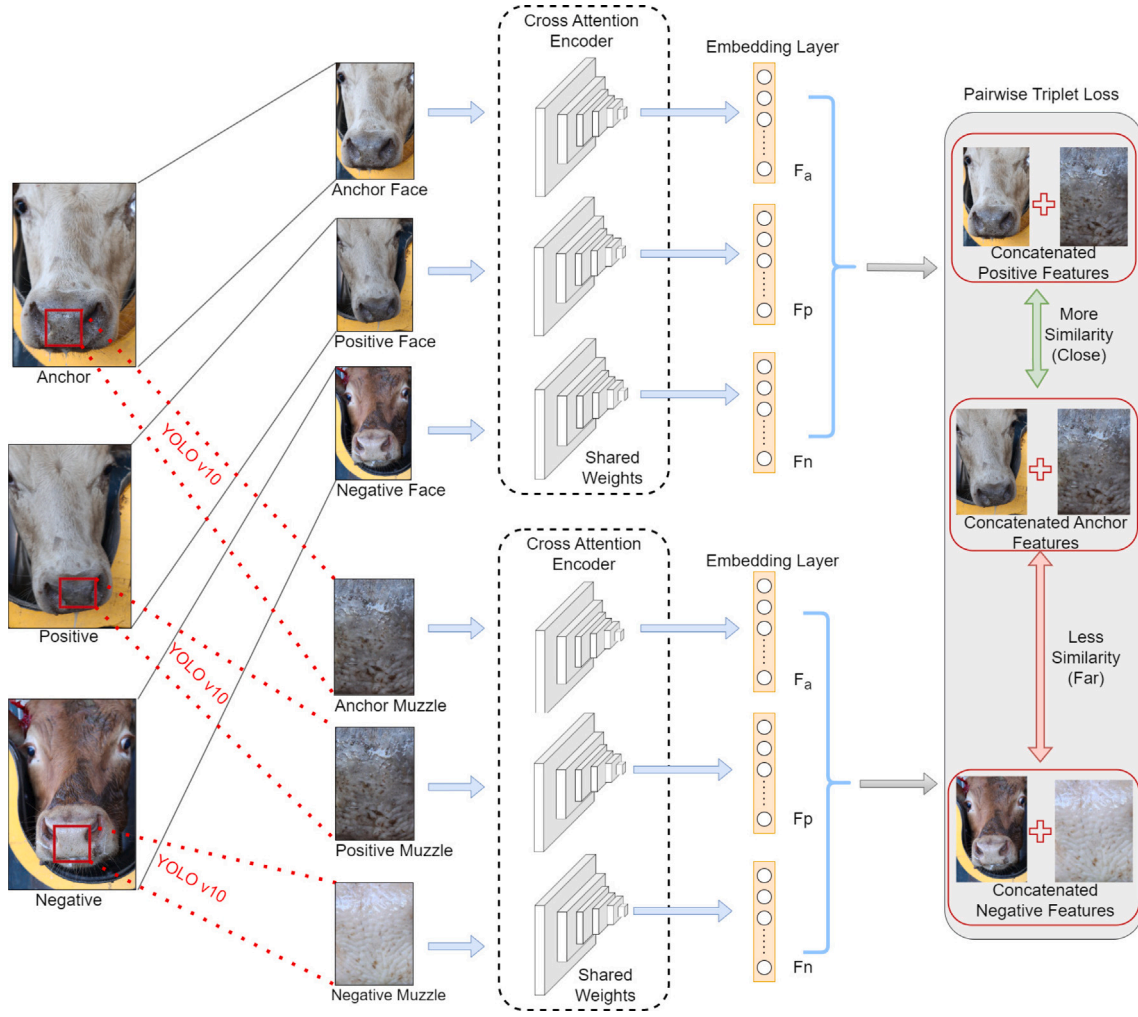


Fig. 4. A workflow of computing pairwise triplet loss considering face and muzzle images.

Unlike the classic triplet loss, which uses a single anchor, positive, and negative image for training, the pairwise triplet loss leverages two anchors, two positives, and two negatives simultaneously. This innovative approach capitalizes on an increased volume of information, intensifying the learning process by refining the model's understanding of pairwise relationships between different features, such as the cattle's face and muzzle. By accommodating more instances in its learning process, the pairwise triplet loss offers the potential for enhanced model performance in cattle verification. The architecture of the pairwise triplet loss is shown in Fig. 4. The Manhattan distance (D_M) is employed as the metric for calculating the similarity between the feature embeddings of the images. The Manhattan distance between two points (x) and (y) in a two-dimensional space is given by:

$$D_M(x, y) = \sum_{i=1}^S |x_i - y_i| \quad (1)$$

Here, x and y represent the feature embeddings of the cattle face or muzzle images, and S represents the embedding size. The Manhattan distance is particularly useful in measuring the absolute differences between corresponding features, making it robust to variations in scale and ensuring that all features contribute equally to the distance calculation. This is crucial for cattle verification, where the model needs to be sensitive to subtle differences in features across different images.

The loss function for the triplet loss is defined as:

$$\mathcal{L}(A, P, N) = \max(D_M(A, P) - D_M(A, N) + m, 0) \quad (2)$$

In this equation, $\mathcal{L}(A, P, N)$ represents the pairwise triplet loss for the triplet (anchor, positive, negative). $D_M(A, P)$ is the Manhattan distance between the anchor pair and the positive pair, while $D_M(A, N)$ is the Manhattan distance between the anchor pair and the negative pair. The margin m is a hyperparameter that defines the minimum difference required between the distances to ensure the model effectively discriminates between the positive and negative pairs. A complete algorithm illustrating the process of cattle verification using the cross-attention encoder-based pairwise triplet loss is shown in Algorithm 1.

The pairwise triplet loss framework offers several advantages that warrant further discussion. Its design of leveraging paired features from both face and muzzle images inherently enhances feature interdependence, resulting in more discriminative embeddings. This dual-feature approach ensures the model can effectively differentiate even between cattle with similar appearances. Furthermore, by utilizing the Manhattan distance, the model is particularly well-suited for scenarios involving subtle feature variations. The pairwise mechanism also introduces robustness to minor perturbations in the data, such as small

variations in pose or occlusions, which are common in real-world scenarios. These characteristics collectively improve the overall reliability and applicability of the method in diverse cattle verification settings.

3.4. Cross attention encoder

The Cross Attention Encoder being proposed here comprises seven convolutional layers. These convolutional layers utilize kernel sizes ranging from 3 to 5 and employ the ReLU activation function. After the sixth convolutional layer, a global average pooling layer is employed as a replacement for the flatten layer. Additionally, the network incorporates a Cross Attention Encoder to enhance the integration of features from both face and muzzle images.

$$\text{Cross Attention}(Q_f, K_m, V_m) = \text{softmax}\left(\frac{Q_f K_m^T}{\sqrt{d_k}}\right) V_m \quad (3)$$

Here in Eq. (3), Q_f represents the query matrix derived from the face features. K_m and V_m represent the key and value matrices derived from the muzzle features. d_k is the dimension of the keys. This equation describes how the query from one modality, like face features, interacts with the keys and values from another modality, i.e., muzzle features, to produce the cross-attention output.

The Cross Attention Encoder is particularly advantageous in this context because it employs a multi-head attention mechanism that captures the most relevant inter-modal relationships between face and muzzle images. This targeted approach ensures that the combined feature representation is more discriminative and robust, which is crucial for accurate cattle verification. Unlike standard convolutional models such as ResNet, Inception, or GoogLeNet, which are designed for general image recognition tasks, the Cross Attention Encoder is specifically tailored to handle the intricate and subtle variations in biometric features between different cattle, thereby improving the model's precision and reliability in this domain. The network concludes with a fully connected layer tailored to the number of target classes. The architecture of the proposed model, including the Cross Attention Encoder, is shown in Table 1. This architecture encompasses a total of approximately 0.67 million parameters. The model's optimization is done with the Adam optimizer during training. During feature generation, the Cross Attention Encoder effectively combines and refines features from both the muzzle and face, leading to a more accurate and reliable output.

Cross Attention Encoder has the ability to dynamically focus on informative regions within both modalities significantly enhances the model's performance under challenging conditions, such as partial occlusions or varying lighting. This mechanism ensures that the model emphasizes regions with the highest discriminative power, improving robustness and generalization. Additionally, the compact architecture of the encoder, with just 0.67 million parameters, enables deployment in resource-limited environments without compromising accuracy. The Cross Attention Encoder's tailored design also ensures efficient feature integration, making it a standout solution for complex biometric tasks like cattle verification. These attributes collectively address challenges in traditional methods and reinforce the applicability of the proposed methodology.

3.5. Adam optimization

The Adam optimizer is particularly tailored for enhancing the optimization of stochastic objective functions in the task of cattle verification using pairwise triplet network. Adam dynamically adjusts estimates of lower-order moments, including the exponentially weighted moving averages of the gradient and squared gradient, to ensure efficient operation (Kingma and Ba, 2017). This adaptive approach provides several advantages over other optimizers. Unlike traditional Stochastic Gradient Descent (SGD) (Ruder, 2017), which uses a fixed learning rate, Adam adjusts the learning rate for each parameter

Table 1

Cross attention encoder architecture details.

Layer type	Layer description	Parameters
Input	Face Input, Shape: (128, 128, 3)	0
Input	Muzzle Input, Shape: (128, 128, 3)	0
Conv2D	32 filters, 3 × 3, ReLU	928
Conv2D	32 filters, 3 × 3, ReLU	9248
Conv2D	32 filters, 5 × 5, Strides = 2, Same, ReLU	25,632
Conv2D	64 filters, 3 × 3, ReLU	18,496
Conv2D	64 filters, 3 × 3, ReLU	36,928
Conv2D	64 filters, 5 × 5, Strides = 2, Same, ReLU	102,464
GAP	Pooling	0
Dense	Fully connected, 250 Units	16,250
Cross attention	Multi-Head Attn., Heads = 4, Dim = 64	254,500
Concatenate	Face features + Attention output	0
Total parameters	–	674,392

individually, allowing it to handle the varying requirements of different parts of the model more effectively. This is crucial in complex tasks like cattle verification, where different features may need different learning rates. Additionally, Adam's mechanism for maintaining a moving average of both the gradients and their second moments enables it to handle sparse gradients more effectively than optimizers like Adagrad (Duchi et al., 2011), which can suffer from an aggressive decay in learning rate over time. Moreover, Adam incorporates bias-correction terms that adjust the estimates of the first and second moments during the initial iterations, which is particularly valuable when working with relatively small datasets, as it ensures that the optimizer makes meaningful progress from the early stages of training. This is especially important in tasks like triplet networks, where learning the relationships between different samples quickly is crucial. Other optimizers, such as RMSprop (Ruder, 2017), lack this bias correction, potentially leading to slower convergence during the initial phases of training. Furthermore, Adam is efficient in terms of memory and computation, requiring only first-order gradients while still offering the benefits of higher-order methods through its moment estimates. These benefits, including computation efficiency, memory usage, and robustness in dealing with diagonal gradient rescaling, make Adam an ideal choice for this application.

The key Eqs. (4), (5), and (6) defining the Adam optimizer are as follows:

$$\hat{m}_t = \frac{\beta_1 m_{t-1} + (1 - \beta_1) g_t}{1 - \beta_1^t} \quad (4)$$

$$\hat{v}_t = \frac{\beta_2 v_{t-1} + (1 - \beta_2) g_t^2}{1 - \beta_2^t} \quad (5)$$

$$\theta_t = \theta_{t-1} - \frac{\alpha \hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \quad (6)$$

The Adam optimizer effectively adjusts the model parameters to minimize the Pairwise triplet loss by adaptive learning rates and bias-corrected moment estimates. The gradient g_t represents the error signal from the triplet loss, guiding the updates to the model's weights. Adam's use of exponential moving averages of the gradient m_t and squared gradient v_t ensures smooth and stable updates, accounting for the variability in cattle images. The bias-corrected estimates $\hat{m}_t$ and $\hat{v}_t$ are crucial for making meaningful progress in the early stages of training, while the learning rate α adapts to the specific needs of different model parameters. Adam's small constant ϵ ensures numerical stability, making the optimizer particularly well-suited for handling the complexities of distinguishing between cattle identities based on diverse facial and muzzle features.

4. Experimental results and discussions

This section provides a brief overview of the experimental setup, including the details of dataset partitioning, triplet formation, and the results obtained during the training and testing phases.

Algorithm 1 Algorithm for Cross-Attention Encoder and Pairwise Triplet Loss

```

1: Input: Cattle Data (Face  $f$  and extracted muzzle  $mz$ )
2: Initialize learning rate  $\alpha$ , margin  $m$ , Encoder layers with cross-attention
3:  $n \leftarrow$  Number of training iterations
4: while  $n > 0$  do
5:   Sample a batch of image triplets: Anchor ( $A$ ), Positive ( $P$ ), Negative ( $N$ )
6:   Pairing of Face( $f$ ) with Muzzle( $mz$ ) Images
       Pair Anchor Images: Face  $f_A$  with Muzzle  $mz_A$ 
       Pair Positive Images: Face  $f_P$  with Muzzle  $mz_P$ 
       Pair Negative Images: Face  $f_N$  with Muzzle  $mz_N$ 
7:   Encode paired features with cross-attention:
        $E_A \leftarrow$  CrossAttentionEncoder( $f_A, mz_A$ ) for Anchor
        $E_P \leftarrow$  CrossAttentionEncoder( $f_P, mz_P$ ) for Positive
        $E_N \leftarrow$  CrossAttentionEncoder( $f_N, mz_N$ ) for Negative
8:   Concatenate the encoded features from the face and muzzle:
        $C_A \leftarrow$  Concatenate( $E_A^f, E_A^{mz}$ ) for Anchor
        $C_P \leftarrow$  Concatenate( $E_P^f, E_P^{mz}$ ) for Positive
        $C_N \leftarrow$  Concatenate( $E_N^f, E_N^{mz}$ ) for Negative
9:   Calculate Manhattan Distance ( $D_M$ ) for each concatenated embedding:
        $D_M(A, P) = \|C_A - C_P\|$ 
        $D_M(A, N) = \|C_A - C_N\|$ 
10:  Calculate Pairwise Triplet Loss ( $L$ ):
        $L(A, P, N) = \max(D_M(A, P) - D_M(A, N) + m, 0)$ 
11:  Update model parameters using the calculated loss and learning rate  $\alpha$ 
12:   $n \leftarrow n - 1$ 
13: end while
14: End

```

4.1. Experimental setup

The platform utilized for this project was a Linux-based operating system, driven by AMD Ryzen processors clocked at 2.4 GHz. This system was outfitted with 16 GB of DDR4 RAM and featured a 1 TB hard drive for ample storage capacity. For graphical processing, an NVIDIA GeForce GPU with 4 GB of VRAM was employed, leveraging CUDA technology for acceleration. The deep learning frameworks used in this setup were TensorFlow and Keras, providing a robust environment for model development and training. During the training process, we experimented with various combinations of hyperparameters to ensure optimal performance. After extensive trials, a batch size of 256 was selected, and the optimizer was configured with a learning rate of $(1e-3)$ and an epsilon value of $(1e-01)$, reflecting the best balance between accuracy and convergence. The model training process took approximately 12 h per epoch for a total of 50 epochs, with the learning rate adjusted using a cosine annealing strategy starting at $1e^{-3}$. The Adam optimizer was utilized with parameters $\beta_1 = 0.9$ and $\beta_2 = 0.999$. Data augmentation techniques such as rotation, flipping, brightness adjustment, and cropping were applied to enhance the robustness of the model. The software environment included Python 3.9 with dependent libraries such as NumPy, Matplotlib, scikit-learn, and OpenCV, all running on CUDA version 11.2.

4.2. Dataset partitioning

To create triplet data for cattle verification, we initially utilized a front-facing cattle images dataset of 300 cattle. These images underwent a series of preprocessing steps, including resizing, detecting the muzzle part using the YOLOv10 algorithm, cropping based on coordinates, and applying image augmentation and transformation. The processed images were then organized into three sets: training, validation, and testing. Of the 300 cattle subjects, 225 were allocated to the training set, 25 to the validation set (10% of the training set), and 50 to the testing set.

4.3. Triplet formation

In similarity learning tasks like cattle verification, it is common to have a different number of classes in the training and testing sets, unlike in classification tasks where the number of classes is usually consistent across all sets. Although we initially used a fixed ratio for training and validation, we also performed an ablation study with different subsets of classes and different numbers of images per class, as shown in Table 6. During model training, images were processed in triplets: two images in each triplet depicted the same cattle (anchor and positive), and the third image featured a different cattle (negative). This approach was applied across the entire dataset, resulting in the formation of a large number of triplet pairs. The calculation of the total number of triplet pairs can be generalized using the following equation.

$$\text{Total Triplets} = F \times (N_{face} \times N_{muzzle})^2 \times (F - 1) \quad (7)$$

where, F represents the number of folders (or classes) in the dataset. N_{face} denotes the number of face images in each folder. N_{muzzle} denotes the number of muzzle images in each folder. $F - 1$ accounts for the selection of a different folder for the negative pair.

This equation reflects the process where every combination of face and muzzle images within each folder forms an anchor pair, with a corresponding positive pair selected from the same folder, and a negative pair chosen from a different folder. Once an image is used as an anchor, positive, or negative, it is not reused, ensuring that each triplet is unique. The ratio of positive to negative pairs in this process is inherently 1:1, as each anchor-positive pair is matched with one corresponding negative pair. This ensures a balanced contribution of both similar (positive) and dissimilar (negative) examples to the training process.

4.4. Results analysis

The training results of the YOLO v10 algorithm for muzzle detection are presented in Table 2. The performance was evaluated using precision, recall, mAP50, and mAP50-95. While the mAP50 value remains constant at 0.985 across all epochs, the mAP50-95 metric shows a significant improvement, increasing from 0.79498 at 20 epochs to 0.96464 at 100 epochs. This indicates that the model becomes more capable of accurately detecting muzzles across a broader range of Intersection over Union thresholds as training progresses. Similarly, the precision and recall metrics improve steadily, reaching 0.99247 and 1.0, respectively, at 100 epochs. These results collectively demonstrate the YOLO v10 algorithm's enhanced ability to precisely and reliably detect the muzzle in input images as the training progresses.

The plots in Fig. 5 depict the training and validation loss (left) and accuracy (right) curves over 100 epochs. The loss curves show a consistent decline for both training and validation sets, with minimal fluctuations after approximately 80 epochs. Specifically, the training loss starts at 0.18 and stabilizes below 0.01 by epoch 100, while the validation loss exhibits a similar trend, stabilizing around 0.005. This indicates that the model effectively learns and converges by this point. Similarly, the accuracy curves for both sets steadily improve during

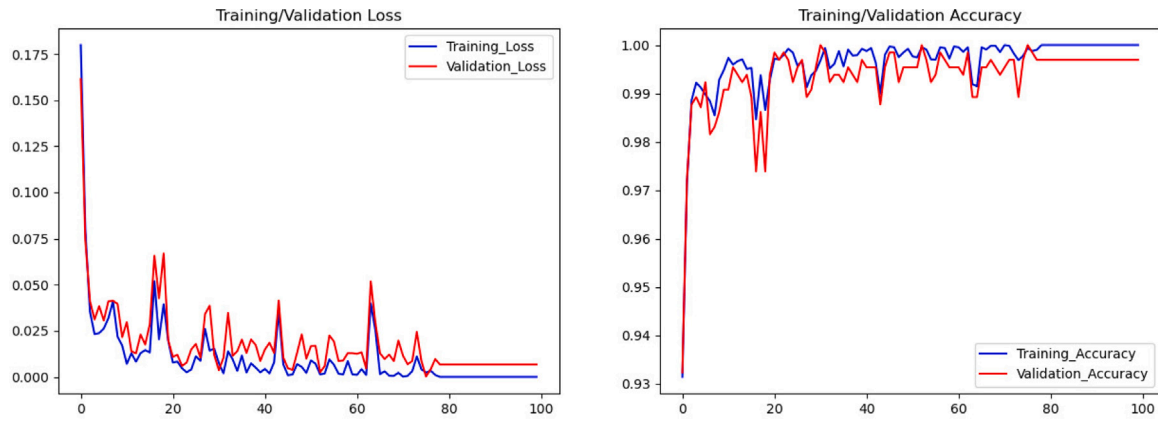


Fig. 5. Training and validation loss vs. Training and validation accuracy curve.

Table 2

Muzzle detection results on YOLO v10 algorithm.

Epochs	Precision	Recall	mAP50	mAP50-95
20	0.96279	0.9849	0.985	0.79498
40	0.97764	0.990	0.985	0.84836
60	0.9717	0.995	0.985	0.9153
80	0.98123	1	0.985	0.94039
100	0.99247	1	0.985	0.96464

the initial epochs, stabilizing around epoch 80, where they achieve near-optimal performance. The training accuracy starts at 93% and reaches 100% by epoch 100, while the validation accuracy begins at 93% and stabilizes at 99.8%. These specific values demonstrate a clear improvement in performance over the training process. The close alignment of the training and validation curves throughout the process suggests minimal overfitting and strong generalization capabilities. These trends validate the conclusion that the model successfully completes training within the specified 100 epochs, reaching a stable and reliable performance level. Furthermore, the quantified improvements in loss and accuracy substantiate the efficacy of the proposed approach in achieving high performance for the given task.

$$\text{False Acceptance Rate (FAR)} = \frac{FP}{FP + TN} \quad (8)$$

$$\text{False Rejection Rate (FRR)} = \frac{FN}{TP + FN} \quad (9)$$

In Eqs. (8), and (9), “True Positive” (TP) signifies when the model correctly predicts a positive outcome, such as accurately identifying a cattle’s anchor and a positive image. Conversely, “True Negative” (TN) denotes when the model accurately predicts a negative outcome, like correctly identifying a cattle’s anchor and a negative image. “False Positive” (FP) occurs when the model predicts a positive outcome when the actual outcome is negative, such as labeling a cattle’s anchor and positive image as similar when they are not. “False Negative” (FN) arises when the model predicts a negative outcome when the actual outcome is positive, such as classifying a cattle’s anchor and a negative image as dissimilar when they are similar.

The training and validation performance of the proposed model is further supported by analyzing the system’s error characteristics, as illustrated in the FAR and FRR plots. These plots provide critical insights into the trade-off between false acceptance and false rejection rates, which are key metrics for evaluating the reliability of the system. Fig. 6 presents the False Acceptance Rate (FAR) and False Rejection Rate (FRR) plots for different margin and threshold values. The graphs illustrate the trade-off between FAR and FRR as the threshold changes. A key observation is the intersection point of the two curves, which represents the threshold where the system achieves equilibrium between minimizing false acceptances and false rejections. This threshold

Table 3

Comparison of the proposed Cross Attention Encoder model’s performance as a feature extractor against various CNN and Transformer-based models. T represents time, E stands for epochs, Inf. denotes inference time, with s and ms indicating seconds and milliseconds, respectively. Newly added models and values are highlighted.

Models	Acc. (%)	Pre. (%)	Rec. (%)	F1 (%)	Params.	T/E	Inf.
VGG	83.1	85.3	85.5	84.6	138M	975 s	250 ms
MobileNet	72.8	77.3	70.7	73.2	3.5M	410 s	50 ms
DenseNet	80.7	78.2	84.3	80.1	7.9M	450 s	60 ms
Inception V3	97.6	95.1	94.3	93.5	22M	580 s	95 ms
ResNet50	86.6	87.2	86.3	85.7	23.5M	610 s	100 ms
RegNetX	77.2	81.6	77.4	79.7	5.6M	430 s	55 ms
RegNetY	88.8	92.5	88.1	88.1	4.3M	420 s	53 ms
Xception	97.1	97.2	96.4	96.2	21M	570 s	92 ms
AlexNet	72.5	76.3	74.4	73.6	61M	750 s	150 ms
SqueezeNet	84.1	87.3	86.4	85.4	1.2M	400 s	40 ms
EfficientNet	86.1	86.2	85.1	83.2	3.5M	420 s	52 ms
ViT	91.5	90.2	89.8	90.4	85M	690 s	110 ms
Swin	92.8	91.8	91.1	91.4	65M	620 s	105 ms
ConvNeXt	92.4	91.7	90.5	91.1	15M	500 s	70 ms
EfficientNetV2	90.8	90.1	89.2	89.6	12M	490 s	65 ms
Proposed	93.6	92.6	91.5	93.1	0.6M	365 s	35 ms

is considered optimal for system performance. For instance, at a margin value of 0.5, the FAR rapidly decreases to nearly 0 by a threshold of 2.5, while the FRR starts increasing significantly after this point. The intersection of the two curves occurs around a threshold of 2, where the FAR and FRR are approximately equal. Similarly, for a margin of 1, the FAR decreases smoothly to 0 by a threshold of around 7, while the FRR increases steadily beyond this threshold, with the intersection point near a threshold of 6. The intersection point of the FAR and FRR curves is significant as it represents a balance between the two error rates. This point is critical in evaluating the model’s reliability, as it indicates the threshold at which the system minimizes both false acceptances and false rejections. By identifying this balance, the system can achieve an optimal trade-off, ensuring robust performance in practical scenarios where minimizing both types of errors is crucial for accurate cattle verification.

The detailed comparison of the proposed Cross Attention Encoder model’s performance against various CNN and Transformer-based architectures in terms of accuracy, precision, recall, F1-score, model size, training time per epoch, and inference time is illustrated in Table 3. Among the compared models, Xception achieves the highest accuracy 97.1% but at the expense of significant computational complexity, with 21 million parameters and an inference time of 92 ms. Similarly, Inception V3 demonstrates a strong balance between accuracy 97.6% and computational efficiency, though it remains computationally more intensive than lightweight models. ResNet50 and VGG, while providing moderate accuracy and F1 scores, are less efficient in terms of training and inference time. The Transformer-based models, including Vision

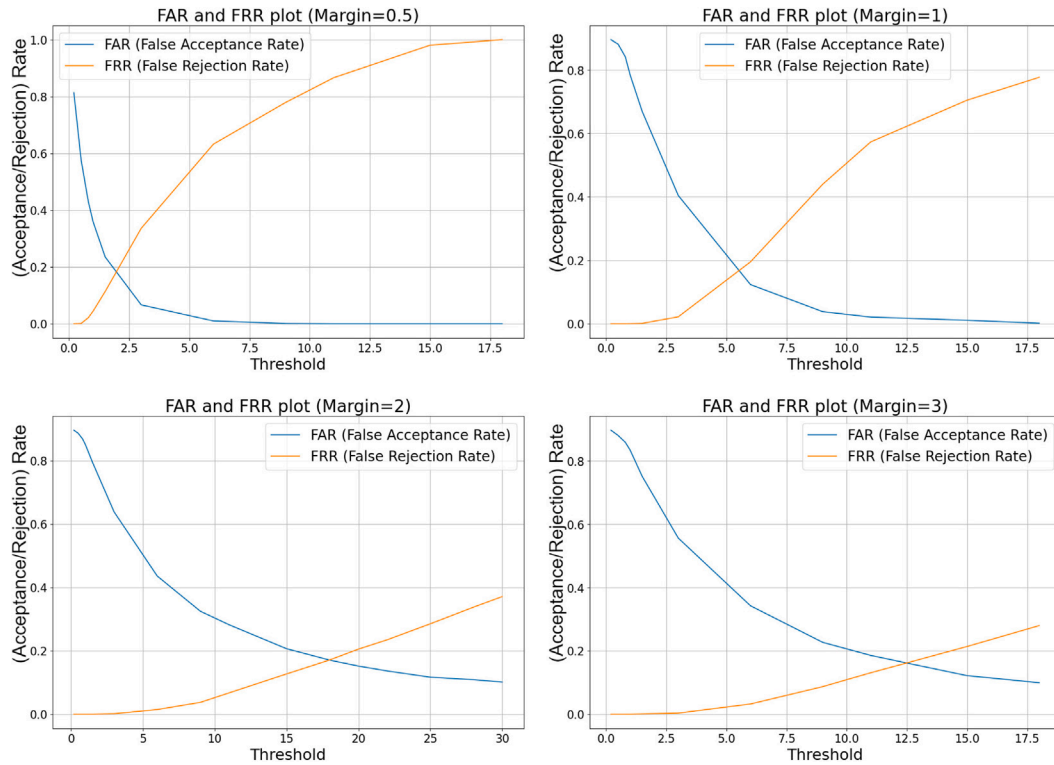


Fig. 6. FAR and FRR curves showing the intersection point, indicating the optimal threshold for improved results.

Transformers (ViT), Swin Transformers, and convolutional models like ConvNeXt, and EfficientNetV2, expand the scope of the evaluation to cover the latest state-of-the-art architectures. Among the transformer models, Swin Transformers demonstrate the accuracy of 92.8% but come with significant computational demands, requiring 65 million parameters and an inference time of 105 ms. Similarly, ViT achieves 91.5% accuracy with 85 million parameters and an inference time of 110 ms, indicating its computational intensity. Models like ConvNeXt and EfficientNetV2 provide slightly lower accuracy of 92.4% and 90.8%, respectively while maintaining a better balance between performance and efficiency. While Swin and ViT offer top-tier accuracy, their high computational and large data requirements limit their applicability in such environments. In contrast, the proposed Cross Attention Encoder model stands out by achieving an exceptional accuracy of 93.6%, precision of 92.6%, recall of 91.5%, and F1-score of 93.1% while requiring significantly fewer parameters of 0.6M. Its computational efficiency is further emphasized by the shortest training time per epoch 365 s and the fastest inference time 35 ms among all compared models. These findings highlight the practicality of the proposed model for real-time and resource-constrained scenarios. Conversely, the proposed model achieves an optimal balance between performance and computational efficiency, making it a uniquely viable solution for livestock management applications.

The classification report presented in Table 4 evaluates the model's performance across different margin and threshold combinations using metrics such as Precision, Recall, Area Under Curve (AUC), False Acceptance Rate (FAR), False Rejection Rate (FRR), and Equal Error Rate (EER). These metrics comprehensively measure the system's ability to correctly classify similar (S) and dissimilar (D) pairs, along with its tendency to produce errors. The results demonstrate that the model achieves its best performance with an AUC of 93.60% at a margin of 1 and a threshold of 6, where a balance between Precision and Recall for both similar and dissimilar pairs is achieved. This setting minimizes both FAR and FRR, resulting in an EER of 0.1437, which underscores its robustness. The observations also reveal distinct trade-offs as the threshold increases: while Precision improves for dissimilar

pairs, Recall for these pairs tends to decrease, reflecting the inherent balance between these metrics. Smaller margins, such as 0.5, exhibit greater sensitivity in detecting dissimilar pairs but are associated with higher error rates, while larger margins like 2 provide more stable performance across thresholds, albeit with slightly lower AUC values. These findings highlight the importance of margin and threshold tuning, as they directly impact model accuracy and error rates. Overall, a margin of 1 with a threshold of 6 emerges as the most effective combination, offering a well-rounded balance between error minimization and classification accuracy for both similar and dissimilar pairs.

The confusion matrices in Fig. 7 illustrate the model's performance at various thresholds for a margin of 1. As the threshold increases from 3 to 11, notable trends emerge in the classification outcomes. True Positives (TP) initially rise but begin to decrease beyond a threshold of 6, while False Positives (FP) decrease until threshold 6 and then increase at higher thresholds. Similarly, True Negatives (TN) increase up to threshold 6 but show a decline at higher thresholds, whereas False Negatives (FN) exhibit the opposite trend, decreasing initially but rising as the threshold increases. These observations indicate that the choice of threshold significantly impacts the balance between correctly and incorrectly classified instances. Threshold 6 stands out as the optimal setting for this dataset, achieving the highest proportion of correctly classified instances (both TP and TN) while minimizing error rates (FP and FN). This threshold provides a balanced trade-off, ensuring reliable classification performance. These findings highlight the importance of carefully selecting the threshold to optimize model accuracy and minimize classification errors.

The accuracy achieved on various publicly available muzzle datasets is compared in Table 5, in relation to the dataset used in our study. To ensure a fair evaluation, we applied consistent preprocessing techniques across all datasets. These datasets vary in terms of the number of classes, image resolution, and the total number of images, which may influence the results. The results demonstrate that our dataset outperforms the others, achieving the highest accuracy of 93.67% for 128-pixel image sizes and 95.40% for 224-pixel image sizes. However, we selected the 93.67% accuracy (128-pixel image size) as the best

Table 4

Classification report for different margin and threshold values, where **S** represents a Similar Match and **D** denotes a Dissimilar Match.

	Threshold	Precision		Recall		AUC	FAR	FRR	EER
		S (0)	D (1)	S (0)	D (1)				
Margin = 0.5	3	0.82	1.00	1.03	0.73	88.56	0.0597	0.3035	0.1817
	6	0.68	1.07	1.10	0.41	75.28	0.0088	0.5715	0.2899
	9	0.62	1.11	1.11	0.24	67.53	0.0007	0.7039	0.3522
	11	0.60	1.11	1.11	0.14	63.10	0.0001	0.7829	0.3911
Margin = 1	3	1.06	0.79	0.68	1.08	87.45	0.3647	0.0196	0.1921
	6	0.91	0.96	0.97	0.89	93.60	0.1114	0.1761	0.1437
	9	0.76	1.04	1.06	0.62	84.13	0.0343	0.3970	0.2155
	11	0.70	1.05	1.08	0.49	77.49	0.0188	0.5175	0.2683
Margin = 2	3	1.11	0.68	0.40	1.11	75.28	0.5759	0.0013	0.2886
	6	1.08	0.76	0.63	1.09	86.35	0.3930	0.0130	0.2031
	9	1.05	0.83	0.75	1.06	90.77	0.2930	0.0333	0.1632
	11	1.01	0.85	0.79	1.03	91.88	0.2544	0.0606	0.1575

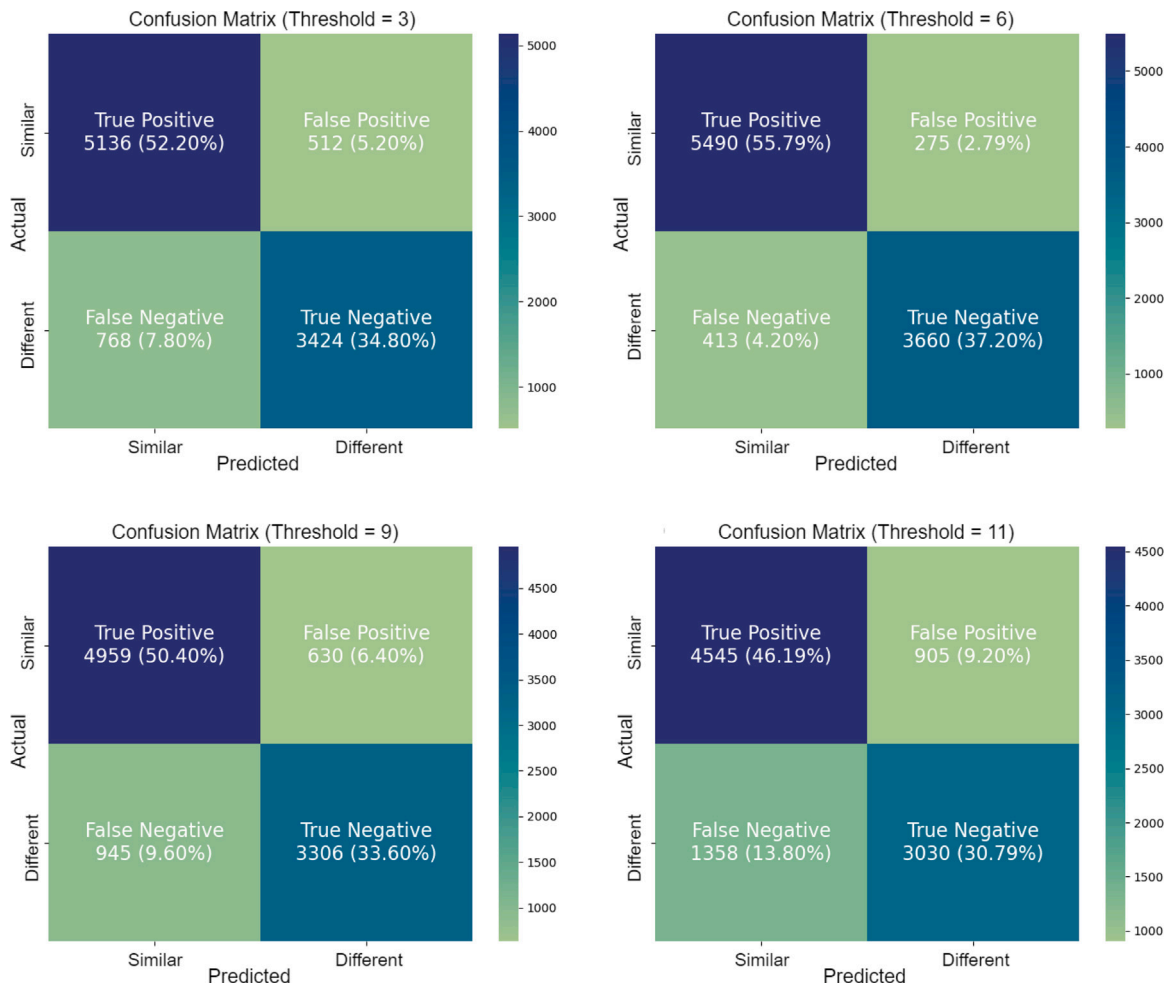


Fig. 7. Confusion matrices at four different threshold values illustrating the model's prediction accuracy when evaluated on individual test samples.

result, as it provides a good balance between accuracy and training time. The training time per epoch for 224-pixel images increases significantly, requiring more hardware resources, which impacts the overall efficiency of the model. In comparison, the dataset from Li et al. reports accuracies of 91.33% and 92.88% for 128-pixel and 224-pixel image sizes, respectively. The dataset from Ahmed et al. shows significantly lower accuracies of 71.40% and 77.19%, respectively. The lower accuracies observed in these datasets can be attributed to several challenges, including occlusions, poor lighting conditions, and environmental factors (such as muzzles obscured by grass), which compromise the quality of the muzzle images. These factors highlight

the importance of using high-quality data for achieving robust cattle identification based on muzzle prints.

The evaluation of the model's performance using k-fold cross-validation highlights its robustness and reliability across varying configurations of the dataset. This method provides valuable insights into how the model generalizes with different numbers of cattle and images per cattle, as well as the influence of the K -fold value on validation accuracy. Table 6 presents the k-fold cross-validation accuracy of the model on our dataset, evaluated across different values of K , numbers of cattle, and images per cattle. The mean validation accuracy (μ) and standard deviation (σ) are calculated over 100 epochs per fold

Table 5

Comparison of accuracy and total training time (Tr. time) across various publicly available cattle datasets. Training time is measured in hours (h).

Dataset	Classes	Total Img.	Img. Res.	Acc. (%)	Tr. time (h)
Li et al. (2022)	230	2480	128 ² 224 ²	91.33 92.88	50 100
Ahmed et al. (2024)	449	4456	128 ² 224 ²	71.40 77.19	70 120
Our study	300	3892	128 ² 224 ²	93.67 95.40	60 110

Table 6

K-fold cross-validation accuracy on our dataset with different K values and different sets of cattle and images.

No. of cattle	Images/cattle	Val. accuracy (%) Mean (μ) $\pm$ Std.Dev (σ)		
		K = 3	K = 4	K = 5
25	5	96 $\pm$ 0.051	95 $\pm$ 0.003	95 $\pm$ 0.001
	10	97 $\pm$ 0.017	97 $\pm$ 0.022	98 $\pm$ 0.001
	15	98 $\pm$ 0.016	98 $\pm$ 0.025	99 $\pm$ 0.001
50	5	97 $\pm$ 0.013	96 $\pm$ 0.004	96 $\pm$ 0.001
	10	98 $\pm$ 0.036	98 $\pm$ 0.003	99 $\pm$ 0.001
	15	99 $\pm$ 0.045	99 $\pm$ 0.004	99 $\pm$ 0.000
100	5	98 $\pm$ 0.006	97 $\pm$ 0.000	97 $\pm$ 0.000
	10	99 $\pm$ 0.034	99 $\pm$ 0.001	99 $\pm$ 0.000
	15	99 $\pm$ 0.001	99 $\pm$ 0.002	99 $\pm$ 0.001

Table 7

Performance comparison of different loss functions for cattle verification.

Loss function	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Cross-entropy loss	88.34	85.67	86.23	85.95
Contrastive loss	91.12	89.34	90.56	89.94
ArcFace loss	93.25	91.67	92.34	92.00
Center loss	89.56	87.34	88.00	87.66
Circle loss	92.89	91.12	91.78	91.45
Pairwise triplet loss	93.67	92.34	92.89	92.61

to provide a detailed understanding of the model's performance and variability. The results reveal a consistent trend. As the number of images per cattle increases, the validation accuracy improves significantly. For instance, when there are only 5 images per cattle, the validation accuracy ranges between 96% and 98%, depending on the K -fold value. In contrast, with 15 images per cattle, the accuracy rises to approximately 99%, accompanied by reduced variability (σ). This trend underscores the positive impact of larger and more diverse training data on model performance. Moreover, the effect of the K -fold value is evident, albeit less pronounced. For a fixed number of images per cattle, higher K -fold values like $K = 5$ generally yield slightly improved accuracy compared to lower K values like $K = 3$. For example, with 10 images per cattle and 50 cattle, the validation accuracy increases from 98% at $K = 3$ to 99% at $K = 5$. The minimal fluctuations in standard deviation across different K -fold values suggest the model performs consistently well regardless of the fold configuration, with the highest variability observed at lower image counts. These observations validate the model's reliability and its ability to generalize effectively across diverse dataset configurations. This comprehensive analysis highlights the importance of a sufficiently large and diverse dataset in achieving high validation accuracy and low variability, providing strong evidence for the model's capability to deliver robust performance across practical scenarios.

The pairwise triplet loss outperforms other loss functions in cattle verification tasks due to its focus on optimizing the relative distances in the embedding space [Table 7](#). Unlike classification-based losses such as cross-entropy, which operate at the level of class probabilities, Pairwise triplet loss ensures that embeddings of the same class (positive pairs) are closer together while those of different classes (negative pairs) are

farther apart by a defined margin. This property is crucial for handling subtle, fine-grained inter-class differences, such as those found in cattle muzzle patterns. Additionally, Pairwise triplet loss is robust to class overlap, a common challenge when dealing with 300 classes in cattle verification. Its ability to generate invariant feature embeddings makes it well-suited to handle variations in lighting, pose, and occlusions, which are typical in real-world datasets. Furthermore, Pairwise triplet loss scales effectively with large datasets by directly working with embeddings rather than relying on individual class predictions, making it a better choice than other loss functions like contrastive or center loss. These strengths collectively make Pairwise triplet loss a powerful and reliable method for achieving high accuracy in cattle verification tasks.

Finally, cattle identification accuracy was compared with other methods, as shown in [Table 8](#). We ensured that the preprocessing was consistent across all cases and used the models proposed by the referenced authors for evaluation. The results were then compared with our proposed method. From the table, it can be observed that the accuracy of the proposed method, which utilizes Pairwise Triplet Loss with face and muzzle features, achieved the highest accuracy of 93.67%. This surpasses the results of several other state-of-the-art methods. For instance, Sanjel et al. achieved an accuracy of 92.21% using the VGGFace model on muzzle features, while Meng et al. obtained 90.90% accuracy using ViT-L and AM-Softmax on face features. Notably, methods utilizing muzzle features generally achieved lower accuracies compared to those using face features, with the exception of the proposed method. This highlights the effectiveness of the Pairwise Triplet Loss approach in enhancing the performance of cattle identification, especially when using both face and muzzle features, which are traditionally more challenging for accurate identification. Although verification and identification are distinct tasks, they share underlying challenges and methodologies, particularly in feature extraction and matching. Since cattle verification is a less explored area with no directly comparable works available, we compared our verification results with the identification accuracy of existing methods.

4.5. Ablation study

The results of various model configurations to isolate the impact of the Cross Attention component are presented in [Table 9](#). The baseline model, which lacks both multiscale processing and attention mechanisms, achieves an accuracy of 85.32% with an inference time of 12.3 ms. Adding only multiscale encoding slightly reduces accuracy to 81.45%, indicating that multiscale processing alone is insufficient for optimal feature representation. Single-modality models also show reduced performance of 78.94% and 80.21%, emphasizing the importance of integrating both face and muzzle modalities. When an attention encoder is introduced, accuracy improves significantly to 88.12%, highlighting the importance of attention in focusing on discriminative features. The proposed model, which combines multiscale encoding, Cross Attention, and dual-modality processing, achieves the best accuracy of 93.67% and reduced inference time of 9.7 ms. This demonstrates that the Cross Attention mechanism synergizes with other components to improve accuracy while optimizing computational efficiency. This underlines the critical role of Cross Attention in the model's architecture and justifies its inclusion as a core component. The multiscale encoder with attention mechanism significantly enhances accuracy compared to baseline configurations. Without the attention mechanism, the accuracy drops to 91.23%, and using a standard encoder without multiscale features further reduces it to 88.45%. This demonstrates that the proposed architecture provides a relative improvement of 4.44% over the best-performing baseline.

The model's robustness under varying lighting, rotations, and occlusions by analyzing accuracy and F1-score is evaluated in [Table 10](#). The model demonstrates strong performance under normal lighting 93.6% accuracy, 93.1% F1-score, while extreme lighting conditions (bright

Table 8
Comparison of accuracy with other state-of-the-art methods applied for cattle identification.

Ref.	Acc. (%)	Feature	Methods
Kumar et al. (2022)	86.20	Muzzle	SURF, IDT
Lee et al. (2023)	88.00	Muzzle	EfficientNetV2
Ruchay et al. (2024)	91.35	Face	VGGFace2
Awad and Hassaballah (2019)	75.70	Muzzle	SIFT, SURF
Shojaeipour et al. (2021)	87.10	Muzzle	ResNet50(Fine Tuned)
Gaber et al. (2016)	71.80	Muzzle	WLD, AdaBoost
Zhang et al. (2023)	89.80	Face	EfficientNet, DeepOtsu
Meng et al. (2023)	90.90	Face	ViT-L, AM-Softmax
Bergman et al. (2024)	90.50	Face	ViT-small
Sanjel et al. (2023)	92.21	Muzzle	VGGFace
Proposed	93.67	Face + Muzzle	Pairwise triplet loss

Table 9
Ablation study on the cross attention component.

Model variant	Multiscale	Attention	Face	Muzzle	Acc. (%)	Inf. (ms)
Baseline	×	×	✓	✓	85.32	12.3
Multiscale encoder	✓	×	✓	✓	81.45	11.8
Single modality	✓	✓	✓	×	78.94	10.5
Single modality	✓	✓	×	✓	80.21	10.8
Attention encoder	×	✓	✓	✓	88.12	13.5
Proposed model	✓	✓	✓	✓	93.67	9.7

Table 10
Impact of data augmentations and transformations on model performance.

Exp	Aug./Trans.	Accuracy (%)	F1-score (%)
Lighting	Bright	91.8	90.6
	Normal	93.6	93.1
	Low	90.4	89.7
Rotations	−30°	92.3	91.4
	15°	92.8	92.2
	30°	91.9	91.1
Occlusion	Small patch	89.6	88.3
	Large patch	87.4	86.0

and low) slightly reduce its effectiveness 91.8% and 90.4% accuracy, respectively. Similarly, rotational variations (−30°, 15°, 30°) show minor impact, with the best results at 15° rotation 92.8% accuracy, 92.2% F1-score, indicating resilience to orientation changes. However, occlusions reveal a significant drop in performance, particularly with larger occlusions 87.4% accuracy, 86.0% F1-score, highlighting the model’s reliance on unobstructed features. These observations confirm the model’s robustness to lighting and rotational changes while identifying sensitivity to occlusions, suggesting potential areas for improvement, such as incorporating occlusion-specific augmentations during training to enhance real-world applicability.

5. Discussion

The proposed Cross-Attention Encoder with Pairwise Triplet Loss function offers significant advancements in the domain of cattle verification. One of its primary advantages lies in its dual-feature approach, which integrates both facial and muzzle features for more robust identification. The Cross-Attention Encoder dynamically emphasizes critical inter-modal relationships, ensuring that the model can extract highly discriminative feature representations. Additionally, the pairwise triplet loss framework effectively leverages multiple positive and negative samples simultaneously, enhancing the model’s ability to learn subtle biometric variations across cattle. The Manhattan distance metric further contributes by ensuring equal sensitivity to all feature dimensions, making the model resilient to small perturbations or variations. Furthermore, the compact design of the proposed architecture, with approximately 0.67 million parameters, makes it computationally efficient, enabling potential deployment in resource-constrained environments.

However, despite these strengths, several limitations need to be addressed to enhance the system’s generalizability and robustness. Data quality issues, such as motion blur caused by cattle movement, remain a critical challenge. Motion blur can obscure fine biometric features, like muzzle patterns, leading to reduced feature clarity and suboptimal verification performance. Another limitation lies in the dataset’s size and diversity. The current dataset may not adequately represent all cattle breeds, age groups, and environmental conditions, potentially limiting the model’s generalization to broader real-world scenarios. Furthermore, complex environmental factors such as extreme lighting, shadow interference, and partial occlusions can obscure key features, reducing the system’s reliability. Lastly, the computational demands of the model, while relatively low, may still present challenges for large-scale deployment in rural or resource-limited settings.

To overcome these limitations, several strategies can be implemented. Addressing motion blur can be achieved through advanced preprocessing techniques, such as employing image restoration models like DeblurGAN or Restormer, which recover sharpness in blurred images. Augmenting the training dataset with synthetically generated blurry images can further improve the model’s robustness to such scenarios. Expanding the dataset to include diverse cattle breeds, varied environmental conditions, and broader geographic coverage would enhance generalizability and reduce the risk of overfitting. To tackle challenges posed by complex environmental conditions, advanced data augmentation techniques can simulate real-world variations during training, allowing the model to learn to recognize features under sub-optimal conditions. Additionally, incorporating attention mechanisms that dynamically focus on clear and informative regions of the input images can mitigate the effects of occlusions and poor lighting. Optimizing the model’s architecture, such as reducing parameters and employing lightweight neural network designs, would enable efficient deployment in resource-constrained environments.

By addressing these limitations and implementing the suggested strategies, the proposed system can become a more robust and scalable solution for smart livestock management, capable of delivering accurate and reliable cattle verification across diverse and challenging real-world scenarios.

6. Conclusion

In this research, we addressed the challenge of cattle verification by proposing a Cross-Attention Encoder with Pairwise Triplet Loss, effectively utilizing both facial and muzzle features. Our model achieved a high testing accuracy of 93.67%, demonstrating its effectiveness in distinguishing individual cattle. Notably, our approach surpassed existing methods, outperforming Sanjel et al. accuracy of 92.21% with VGGFace and Meng et al. accuracy of 90.90% with ViT-L and AM-Softmax, reinforcing the advantage of our feature fusion strategy. Furthermore, the model’s lightweight architecture, with a reduced parameter count of 0.6M, ensures computational efficiency, making it feasible for real-time applications in farms and ranches where resources are limited. Despite these promising results, certain challenges remain. Motion blur caused

by cattle movement during image capture can affect feature clarity, impacting verification accuracy. Additionally, the current dataset, while effective, would benefit from further expansion to include more diverse breeds and environmental conditions, enhancing generalization. Future research will focus on refining model structure to further reduce inference time while maintaining high accuracy. Investigating advanced multimodal fusion strategies could also improve feature integration, leading to even more robust verification. Moreover, leveraging transfer learning and deploying the model on edge AI devices will be explored to facilitate real-time cattle identification in practical settings. This study highlights the potential of integrating attention-based feature learning for cattle verification, offering a scalable and automated solution to livestock identification that can significantly enhance precision livestock management.

CRedit authorship contribution statement

Niraj Kumar: Writing – original draft, Methodology. **Anshul Sharma:** Writing – review & editing. **Abhinav Kumar:** Writing – review & editing, Resources, Funding acquisition. **Rishav Singh:** Writing – review & editing, Data curation. **Sanjay Kumar Singh:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We gratefully acknowledge the support from the PARAM Shivay Facility under the National Supercomputing Mission, Government of India, at IIT (BHU), Varanasi, and the funding from the NASF project ‘Artificial Intelligence & IoT-based Smart Vet Ecosystem for Animal Health, Patient Care, and Precision Livestock Farming’ (Grant No. NASF/PA-9028/2022-23).

Data availability

Publicly available dataset is used for evaluation.

References

- Ahmed, S.U., Frnda, J., Waqas, M., Khan, M.H., 2024. Dataset of cattle biometrics through muzzle images. *Data Brief* 53, 110125. <http://dx.doi.org/10.1016/j.dib.2024.110125>.
- Awad, A.I., Hassaballah, M., 2019. Bag-of-visual-words for cattle identification from muzzle print images. *Appl. Sci.* 9 (22), 4914. <http://dx.doi.org/10.3390/app9224914>.
- Bergman, N., Yitzhaky, Y., Halachmi, I., 2024. Biometric identification of dairy cows via real-time facial recognition. *Animal* 18 (3), 101079.
- Bhole, Amey, Udmale, Sandeep S., Falzon, Owen, Azzopardi, George, 2022. Corf3d contour maps with application to holstein cattle recognition from rgb and thermal images. *Expert Syst. Appl.* 192 (C).
- Bochkovskiy, Alexey, Wang, Chien-Yao, Liao, Hong-Yuan Mark, 2020. YOLOv4: Optimal speed and accuracy of object detection.
- Dubourvieux, Fabian, Lapouge, Guillaume, Loesch, Angélique, Luvison, Bertrand, Audigier, Romaric, 2023. Cumulative unsupervised multi-domain adaptation for holstein cattle re-identification. *Artif. Intell. Agric.* 10, 46–60.
- Duchi, John, Hazan, Elad, Singer, Yoram, 2011. Adaptive subgradient methods for online learning and stochastic optimization. *J. Mach. Learn. Res.* 12, 2121–2159.
- Dulal, Rabin, Zheng, Lihong, Kabir, Muhammad Ashad, 2025. MHAFF: Multi-head attention feature fusion of CNN and transformer for cattle identification. *arXiv preprint*.
- Duong, Chi Nhan, Quach, Kha Gia, Le, Ngan, Nguyen, Nghia, Luu, Khoa, 2018. Mobiface: A lightweight deep learning face recognition on mobile devices. *CoRR*, abs/1811.11080.
- Gaber, T., Tharwat, A., Hassanien, A.E., Snasel, V., 2016. Biometric cattle identification approach based on Weber's local descriptor and AdaBoost classifier. *Comput. Electron. Agric.* 122, 55–66.
- Girshick, Ross, Donahue, Jeff, Darrell, Trevor, Malik, Jitendra, 2014. Rich feature hierarchies for accurate object detection and semantic segmentation.
- Hoffer, Elad, Ailon, Nir, 2018. Deep metric learning using triplet network.
- Kingma, Diederik P., Ba, Jimmy, 2017. Adam: A method for stochastic optimization.
- Kuhl, Hjalmar S., Burghardt, Tilo, 2013. Animal biometrics: quantifying and detecting phenotypic appearance. *Trends Ecol. Evol.* 28 (7), 432–441.
- Kumar, S., Chaube, M.K., Kumar, S., 2022. Secure and sustainable framework for cattle recognition using wireless multimedia networks and machine learning techniques. *IEEE Trans. Sustain. Comput.* 7 (3), 696–708.
- Kumar, Santosh, Tiwari, Shrikant, Singh, Sanjay Kumar, 2016. Face recognition of cattle: Can it be done? *Proceedings of the national academy of sciences. India Sect. A: Phys. Sci.* 86, 137–148.
- Lee, T., Na, Y., Kim, B.G., Lee, S., Choi, Y., 2023. Identification of individual hanwoo cattle by muzzle pattern images through deep learning. *Animals* (ISSN: 2076-2615) 13 (18), 2856. <http://dx.doi.org/10.3390/ani13182856>, Available at: <https://www.mdpi.com/2076-2615/13/18/2856>. PMID: 37760256.
- Li, G., Erickson, G.E., Xiong, Y., 2022. Individual beef cattle identification using muzzle images and deep learning techniques. *Anim. (Basel)* 12 (11), 1453. <http://dx.doi.org/10.3390/ani12111453>, PMID: 35681917. PMCID: PMC9179917.
- Li, Dongxu, Li, Baoshan, Li, Qi, Wang, Yueming, Yang, Mei, Han, Mingshuo, 2024. Cattle identification based on multiple feature decision layer fusion. *Sci. Rep.*.
- Lin, Tsung-Yi, Goyal, Priya, Girshick, Ross, He, Kaiming, Dollar, Piotr, 2018. Focal loss for dense object detection.
- Liu, Decheng, Gao, Xinbo, Wang, Nannan, Li, Jie, Peng, Chunlei, 2020. Coupled attribute learning for heterogeneous face recognition. *IEEE Trans. Neural Netw. Learn. Syst.* 31 (11), 4699–4712.
- Liu, Shu, Qi, Lu, Qin, Haifang, Shi, Jianping, Jia, Jiaya, 2018. Path aggregation network for instance segmentation.
- Lu, Yongzheng, Weng, Zhi, Zheng, Zhiqiang, Zhang, Yong, Gong, Caili, 2023. Algorithm for cattle identification based on locating key area. *Expert Syst. Appl.* 228, 120365.
- Ma, Chunmei, Sun, Huazhi, Zhu, Jinqi, Zhang, Long, Wang, Bojue, Wu, Donghao, Sun, Jingwei, 2021. Normalized maximal margin loss for open-set image classification. *IEEE Access* 9, 54276–54285.
- Mahato, Shubhangi, Neethirajan, Suresh, 2024. Integrating artificial intelligence in dairy farm management biometric facial recognition for cows. *Inf. Process. Agric.*.
- Meng, Yao, Yoon, Sook, Han, Shujie, Fuentes, Alvaro, Park, Jongbin, Jeong, Yongchae, Park, Dong Sun, 2023. Improving known-unknown cattle's face recognition for smart livestock farm management. *Animals* 13 (22), 3588.
- Noviyanto, Ary, Arymurthy, Aniati Murni, 2013. Beef cattle recognition based on muzzle pattern using a matching refinement technique in the sift method. *Comput. Electron. Agric.* 99, 77–84.
- Pinto, Joao Ribeiro, Correia, Miguel V., Cardoso, Jaime S., 2021. Secure triplet loss: Achieving cancelability and non-linkability in end-to-end deep biometrics. *IEEE Trans. Biom. Behav. Identity Sci.* 3 (2), 180–189.
- Redmon, J., Divvala, S., Girshick, R., Farhadi, A., 2016. You only look once: Unified, real-time object detection. *arXiv preprint*.
- Reis, D., Kupec, J., Hong, J., Daoudi, A., 2024. Real-time flying object detection with YOLOv8.
- Ren, S., He, K., Girshick, R., Sun, J., 2016. Faster R-CNN: Towards real-time object detection with region proposal networks. *arXiv preprint arXiv:1506.01497*.
- Ren, Guoyin, Lu, Xiaoqi, Li, Yuhao, 2021. A cross-camera multi-face tracking system based on double triplet networks. *IEEE Access* 9, 43759–43774.
- Ren, Fuji, Xue, Siyuan, 2020. Intention detection based on siamese neural network with triplet loss. *IEEE Access* 8, 82242–82254.
- Ruchay, A., Kolpakov, V., Guo, H., Pezzuolo, A., 2024. On-barn cattle facial recognition using deep transfer learning and data augmentation. *Comput. Electron. Agric.* 225, 109306. <http://dx.doi.org/10.1016/j.compag.2024.109306>.
- Ruder, Sebastian, 2017. An overview of gradient descent optimization algorithms. *arXiv preprint arXiv:1609.04747*.
- Sanjel, A., Khanal, B., Rivas, P., Speegle, G., 2023. Non-invasive muzzle matching for cattle identification using deep learning. In: 2023 Congress in Computer Science, Computer Engineering, & Applied Computing. CSCE, pp. 1998–2002.
- Schroff, F., Kalenichenko, D., Philbin, J., 2015. FaceNet: A unified embedding for face recognition and clustering. In: 2015 IEEE Conference on Computer Vision and Pattern Recognition. CVPR, IEEE.
- Shen, Wen, Hu, Hui, Dai, Baolong, et al., 2020. Individual identification of dairy cows based on convolutional neural networks. *Multimedia Tools Appl.* 79, 14711–14724.
- Shojaeipour, Ali, Greg, Falzon, Paul, Kwan, Nooshin, Hadavi, Cowley Frances, C., David, Paul, 2021. Automated muzzle detection and biometric recognition via few-shot deep transfer learning of mixed breed cattle. *Agronomy* 11 (11).
- Wang, Chien-Yao, Bochkovskiy, Alexey, Liao, Hong-Yuan Mark, 2022. YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors.
- Wang, A., Chen, H., Liu, L., Chen, K., Lin, Z., Han, J., Ding, G., 2024. YOLOv10: Real-time end-to-end object detection. *arXiv preprint*.
- Xie, Weicheng, Wu, Haoqian, Tian, Yi, Bai, Mengchao, Shen, Linlin, 2022. Triplet loss with multistage outlier suppression and class-pair margins for facial expression recognition. *IEEE Trans. Circuits Syst. Video Technol.* 32 (2), 690–703.
- Xu, Xingshi, Deng, Hongxing, Wang, Yunfei, Zhang, Shujin, Song, Huaibo, 2024. Boosting cattle face recognition under uncontrolled scenes by embedding enhancement and optimization. *Appl. Soft Comput.* 164, 111951.

- Xu, Beibei, Wang, Wensheng, Guo, Leifeng, Chen, Guipeng, Li, Yongfeng, Cao, Zhen, Wu, Saisai, 2022. Cattlefacenet: A cattle face recognition approach based on retinaface and arcface loss. *Comput. Electron. Agric.* 193 (C).
- Yu, Jun, Zhu, Chaoyang, Zhang, Jian, Huang, Qingming, Tao, Dacheng, 2020. Spatial pyramid-enhanced netvlad with weighted triplet loss for place recognition. *IEEE Trans. Neural Netw. Learn. Syst.* 31 (2), 661–674.
- Zhang, Ruihong, Ji, Jiangtao, Zhao, Kaixuan, Wang, Jinjin, Zhang, Meng, Wang, Meijia, 2023. A cascaded individual cow identification method based on DeepOtsu and EfficientNet. *Agriculture* 13 (2), 279.
- Zhao, Jian-Min, Lian, Qiu-Sheng, 2022. Compact loss for visual recognition of cattle in the wild. *Comput. Electron. Agric.* 195 (C).